Generated Notes

Here are notes explaining operating systems, based on the provided material:

OPERATING SYSTEM BASICS

What is an Operating System?

* An Operating System (OS) is a program that acts as a bridge between a computer user and the computer's hardware.

* It is a collection of system programs that work together to manage how a computer system operates.

* **Examples:** UNIX, Mach, MS-DOS, MS-Windows, Windows/NT, OS/2, MacOS, VMS, MVS, VM.

Goals of an Operating System

* To run user programs and make solving user problems easier.

* To make the computer system convenient and simple to use.

* To use the computer hardware efficiently.

Computer System Components

1. **Hardware:** Provides the basic computing power, including:

* CPU (Central Processing Unit)

* Memory

* I/O devices (Input/Output devices like keyboard, mouse, printer)

2. **Operating System:** Controls and coordinates how the hardware is used among different programs and users.

VIRTUAL MACHINES

What is a Virtual Machine?

* A virtual machine creates an environment that looks and acts exactly like the underlying physical hardware.

* The operating system makes it seem like multiple programs (processes) are each running on their own dedicated processor with their own separate (virtual) memory.

* The actual resources of the physical computer are shared to create these virtual machines.

How the Illusion is Created

* **CPU Scheduling:** Makes it appear that each user has their own dedicated processor.

* **Spooling and File System:** Can make it seem like there are virtual card readers and virtual line printers.

* **User Terminal:** A regular time-sharing terminal can act as the control console for a virtual machine.

System Models

* There are two main ways to think about computer systems:

* Non-virtual Machine (traditional system)

* Virtual Machine (system where resources are virtualized)

Advantages of Virtual Machines

* The virtual-machine idea offers complete protection between different virtual machines.

CHALLENGES IN OPERATING SYSTEMS

Dining Philosophers Problem

* This is a classic problem used to illustrate challenges in managing shared resources.

* **Scenario:** Five philosophers sit around a circular table with a bowl of rice in the middle. There are five single chopsticks, one between each pair of philosophers.

* **Behavior:**

* Philosophers spend time thinking.

* When hungry, a philosopher tries to pick up the two chopsticks closest to them (one on their left, one on their right).

* They can only pick up one chopstick at a time.

* They cannot pick up a chopstick that another philosopher is already holding.

* Once a hungry philosopher has both chopsticks, they eat without letting go.

* When finished eating, they put down both chopsticks and return to thinking.

Critical Section Problem

* This problem arises when multiple processes (programs) need to access a shared piece of code or data, called a "critical section."

* The main goal is to ensure that only one process can be inside its critical section at any given time.

Conditions for a Solution to the Critical Section Problem

1. **Mutual Exclusion:** If one process is using the critical section, no other process can enter its critical section.

2. **Progress:** If no process is currently in the critical section, and some processes want to enter, then only those processes that are not busy with other tasks can decide which one enters next. This decision cannot be delayed forever.

3. **Bounded Waiting:** After a process asks to enter its critical section, there is a limit to how many other processes can enter their critical sections before that process gets its turn.

Peterson’s Solution

* Peterson's solution is a way to solve the critical section problem using software.

* It is designed for two processes, for example, Process P0 and Process P1.